

MATT HALL PhD

mattdanhall.com
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Software engineer with a passion for innovation, research and problem-solving. Currently involved in the re-imagining of a critical but ageing healthcare system that desperately requires cloud migration and other modernisation. Historical expertise in designing, developing, evaluating and presenting control algorithms using custom built simulation software. Proven track record of effectively communicating complex and varied concepts to audiences of differing backgrounds. Seeking a new and engaging challenge in an ambitious, mission-driven company that is aiming to foster novelty and innovation at its core.

COMPETENCIES & TECH SKILLS

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Python • AWS • Terraform • Git • CI/CD • Cloud Architecture • API Development • TDD • Web Technologies • Algorithm Design • Agile Methodologies • *nix • Shell Scripting • Modular & Swarm Robotics • LaTeX • Technical Writing/Presentation • Data Analysis

Rust • HTML • CSS • Java • C/C++ • MATLAB • Gherkin • TypeScript • Docker • VHDL

EXPERIENCE

Software Engineer – BJSS

JUN 2022 – PRESENT

Joined via intensive 4 week engineering academy program, using multiple languages to gain practical experience with cloud architecture, agile approach, test-driven development and more. Immediately moved onto long-running client project, migrating a healthcare system to AWS using Python, Terraform and Typescript. Member of Data Assurance Team, requiring attention to detail and adaptability to client needs. Enlisted to join Transition Team, responsible for go-live migration. This requires broad knowledge and ability to work under pressure while maintaining high standards.

Postgraduate Researcher – Control & Systems Engineering

SEP 2017 – OCT 2022

Developed and verified novel control algorithms for the control of modular, multi-robot systems. Created custom simulation and evaluation software using Python to verify work. Responsible for the entire research cycle, from idea generation to scientific reporting. Collaborated with academics throughout the process, including as a mentor to other research students.

Peer Reviewer – Natural Robotics Lab

SEP 2017 – SEP 2021

Reviewed submitted works for a number of highly regarded international conferences, such as AAMAS, ANTS, DARS, IROS, MRS, RSS and TAROS. Necessary to keep up to date with state-of-the-art work. Utilised my ability to disseminate information and assess the quality of written works.

Graduate Teaching Assistant – The University of Sheffield

JAN 2018 – DEC 2020

Assisted in the planning and running of laboratory-based teaching and seminar sessions for Master's and Bachelor's students. Required in-depth knowledge of the material and the ability to clearly communicate to answer student questions. Organisational skills necessary to effectively plan sessions for multiple groups.

Tourism Advisor – Use-it Brussels

DEC 2016 – AUG 2017

Part-time work within a sustainable tourism project while residing in Belgium, offering tourist information and running tours of the lesser-known parts of Brussels. A great opportunity to thoroughly learn about the city and its culture, as well as improving my interpersonal skills.

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EDUCATION

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PhD - *Automatic Control and Systems Engineering*

2017 - 2022

The University of Sheffield

THESIS: “Active Subtraction: A Viable Method of Self-Reconfiguration for Modular Robotic Systems”

OVERVIEW: The creation, verification and demonstration of a number of novel control algorithms to guarantee autonomous reconfiguration of multi-robot modular systems, enabling modules to remove themselves from a given starting configuration to form any desired shape.

MEng (Master of Engr.) - *Electrical and Electronics Engineering*

2012 - 2016

Lancaster University - **1st Class Hons**

A-Levels - *Maths (A*) Further Maths (A) Physics (B)*

2010 - 2012

Leeds, Abbey Grange Academy Sixth Form - **Governor’s Award for Outstanding Results**

PUBLICATIONS

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M. D. Hall, A. Özdemir, R. Groß, “Self-Reconfiguration in Two-Dimensions via Active Subtraction with Modular Robots”, in *Proc. Robotics: Science and Systems XVI (RSS)*, RSS Foundation, 2020. (Presented virtually: https://youtu.be/RgGi_mc4t0Q)

A. Özdemir, M. Gauci, A. Kolling, M. D. Hall, R. Groß, “Spatial Coverage Without Computation”, in *Proc. 2019 International Conference on Robotics and Automation (ICRA)*, IEEE, 2019, pages 9674–9680. (Presented in person)

ACADEMIC AWARDS

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EPSRC Doctoral Training Partnership National Productivity Scholarship for course and maintenance fees • University Postgraduate Research Student Publication Scholarship for 3 months funding towards journal publication • Best Oral Presentation at Departmental Symposium • Best Poster Presentation at Faculty Researcher Symposium

FURTHER INTERESTS

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PROGRAMMING: Designed and created my [personal website](#) using the **Hugo** framework. Released a small number of customisation apps for KLWP on **Android**. Selection of other projects available on my [GitLab](#).

MUSIC: Bassist and guitarist. Experience in Ableton music production. Regular guest [radio DJ](#) for “Clear A Space” on Gaga Radio.

TRAVEL: Enjoy learning about cultures and being immersed in new cities. Lived and worked in Brussels following Master’s degree completion.

FILM: Keen interest in film. **Written**, **shot** and **edited** amateur films with family and friends.

KORFBALL: Assistant coach and 2nd team captain for Lancaster Ligers Korfball. Social secretary for the club during final year.

OTHER: Food and cooking, games (board or video), craft beer.

REFERENCES AVAILABLE UPON REQUEST

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